

# SYSTEM DESIGN

### 3.1 System Architecture

Electroencephalography (EEG) is a powerful tool used to measure electrical activity in the brain, making it highly relevant for applications such as diagnosing neurological disorders, mental health assessments, and cognitive research. In the context of Major Depressive Disorder (MDD) or other cognitive conditions, EEG data can be leveraged in a structured machine learning pipeline to enhance early detection and analysis.

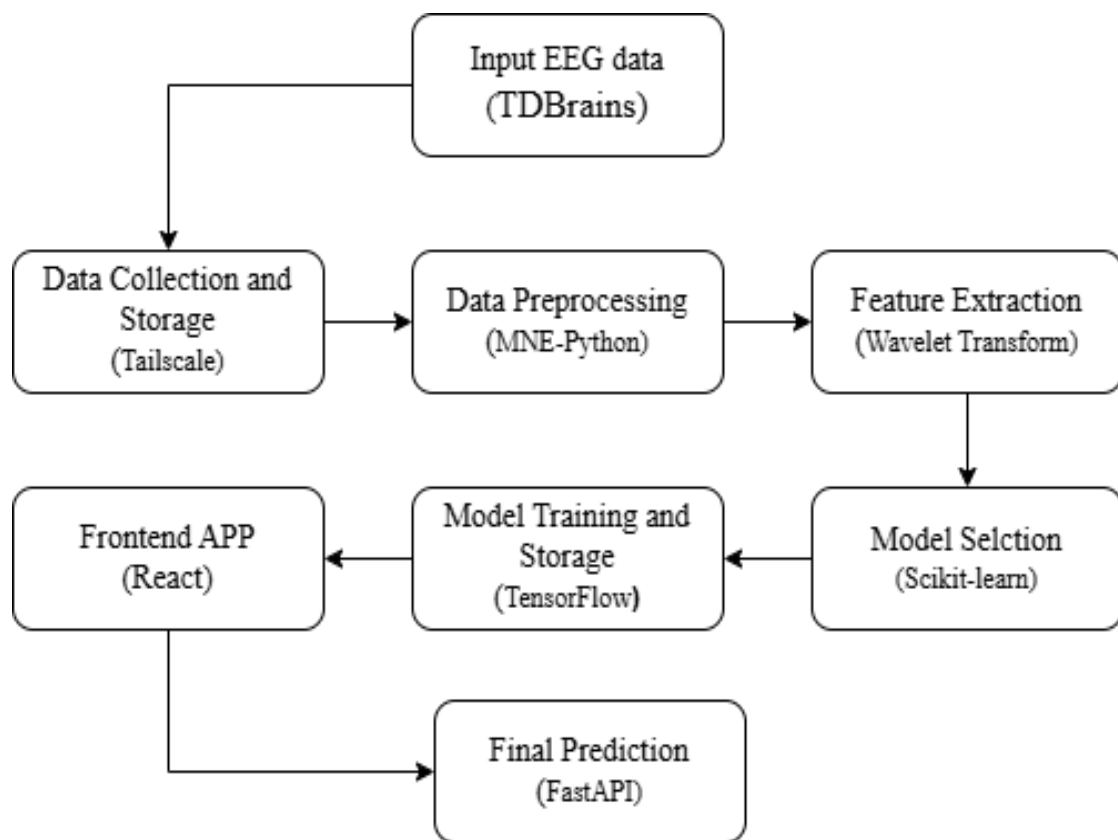


Fig no.3.1 Architectural diagram

The above Fig 3.1 provides a comprehensive explanation of the different stages involved in an EEG-based machine learning system, focusing on data processing, model training, and implementation through a user-friendly frontend application.

#### 1. Input EEG Data

The system is acquiring raw EEG data from patients or subjects. EEG signals are captured using electrodes placed on the scalp, recording electrical brain activity over time. The signals are typically recorded in multiple

channels corresponding to different brain regions. This raw data forms the foundation for further processing and analysis. This data is obtained from TDBrains clinics dataset.

## 2. Data Collection and Storage

Once acquired, the EEG data needs to be systematically collected and stored. This step ensures that the data remains organized, accessible, and formatted correctly for future processing. The data is usually stored in structured databases, cloud storage, or local servers with proper indexing and metadata to facilitate efficient.

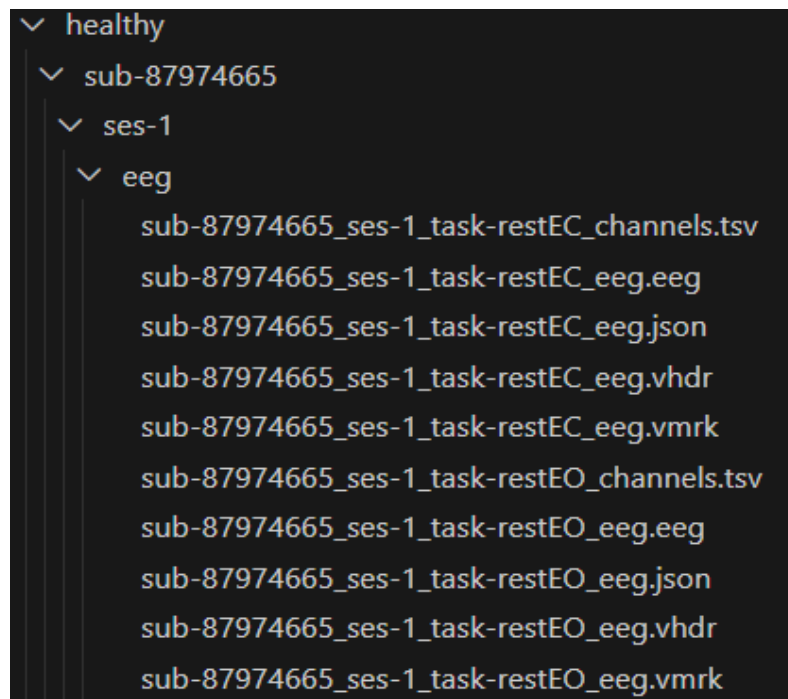


Fig no. 3.2 Hierarchical structure of data

The above Fig 3.2 illustrates hierarchical structure of data organizes data in a tree-like format with parent-child relationships, where each parent can have multiple children, but each child has only one parent.

TABLE 3.1. Description of Files in Dataset

| File Name | Description                                                                                       |
|-----------|---------------------------------------------------------------------------------------------------|
| .eeg      | The raw EEG data (binary format).                                                                 |
| .vhdr     | The header file containing metadata about the EEG recording (sampling rate, channel names, etc.). |
| .vmrk     | The marker file that contains event markers (stimuli, blinks, artifacts, etc.).                   |
| .json     | Metadata file containing subject-specific details.                                                |
| .tsv      | Tab-separated file with additional channel-specific metadata.                                     |

This table 3.1 describes EEG data files, where. Eeg stores raw brain signals, vhdr provides metadata like sampling rate, and. vmrk logs event markers such as stimuli. The .Json file contains subject-specific details, while. tsv includes channel metadata like electrode positions. together, these files ensure accurate EEG data processing and analysis.

### 3. Data Preprocessing

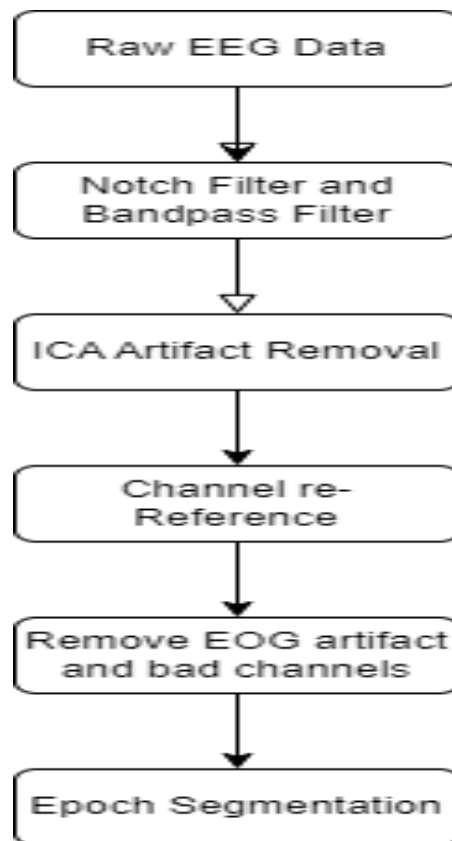


Fig no. 3.3 Pipeline for preprocessing

The above Fig 3.3 illustrates the systematic preprocessing pipeline used for EEG signal processing to enhance data quality and remove unwanted noise and artifacts. Preprocessing EEG signals is a critical step to ensure that the data used for analysis and classification is free from noise, artifacts, and irrelevant information. The raw EEG signals are often contaminated with various sources of noise, such as power line interference, eye movements, and muscle activity, which can significantly affect the accuracy of the final results. A structured approach is followed to improve the signal quality and ensure that meaningful neural activity is preserved.

The EEG signals are recorded from subjects using an EEG acquisition system equipped with multiple electrodes placed on the scalp. The placement of electrodes follows the 10-20 international system to ensure standardized positioning across participants. Each electrode captures electrical activity generated by neurons, but this raw signal also includes unwanted noise from external and physiological sources.

#### 4. Feature Extraction

After preprocessing, the next step involves extracting meaningful features from the EEG data. These features represent the most relevant characteristics of brain activity, which help in distinguishing different mental states or conditions. Feature extraction methods can include time-domain analysis (e.g., statistical moments), frequency-domain analysis (e.g., power spectral density), and time-frequency representations (e.g., wavelet transforms). Selecting the right set of features improves model accuracy and interpretability.

Feature extraction in EEG analysis is a crucial step that involves transforming raw or preprocessed signals into meaningful representations that capture the most relevant aspects of brain activity. This process enhances the ability of machine learning models to classify different mental states, cognitive processes, or neurological conditions. Various feature extraction techniques are applied depending on the type of information required.

Time-domain features provide statistical descriptions of the EEG signal by analyzing its amplitude variations over time. Commonly used features include the mean, variance, skewness, kurtosis, and peak-to-peak amplitude, which help quantify signal fluctuations. Additionally, Hjorth parameters, such as activity, mobility, and complexity, offer insights into the signal's dynamic properties. These features are particularly useful for applications where temporal variations play a crucial role, such as event-related potential (ERP) analysis.

TABLE 3.2: Extracted EEG Features and their Description

| Feature Type     | Feature Description    |                                                     |
|------------------|------------------------|-----------------------------------------------------|
|                  | Feature Name           | Description                                         |
| Time-Domain      | Mean Amplitude         | Average signal value over time.                     |
|                  | Variance               | Measures signal fluctuation intensity.              |
|                  | Skewness & Kurtosis    | Describes the shape of the EEG signal distribution. |
| Frequency-Domain | FFT Power Spectrum     | Decomposes EEG into frequency components.           |
|                  | PSD (Welch's Method)   | Estimates power spectral density for EEG signals.   |
|                  | Delta Power (0.5–4 Hz) | Brain state during deep sleep and unconsciousness.  |

The above table 3.2 illustrates various EEG feature extraction methods categorized into time-domain, frequency-domain, and time-frequency features, each providing valuable insights into brain activity. Time-domain features

analyse the amplitude variations of the EEG signal over time. The mean amplitude represents the average signal value, while variance measures the intensity of signal fluctuations. Skewness and kurtosis describe the shape of the EEG signal distribution, helping to understand data spread and detect extreme values.

## 5. Model Selection

The extracted features serve as the foundation for selecting a suitable machine learning model for EEG classification. The choice of model depends on various factors, including the complexity of EEG signal patterns, the size of the dataset, and the desired level of accuracy. To achieve optimal performance, model selection involves rigorous evaluation processes such as hyperparameter tuning, cross-validation, and performance metric analysis. Different models are assessed based on their ability to capture temporal and spectral characteristics of EEG signals while maintaining generalization across different subjects and conditions. The final selection is guided by a balance between computational efficiency and classification accuracy, ensuring reliable and interpretable results.

Table 3.3: Models used for classification

| <b>Model</b>                 | <b>Model Decryption</b>  |                                                                                                               |
|------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------|
|                              | <b><i>Model Type</i></b> | <b><i>Model features</i></b>                                                                                  |
| Logistic Regression (LR)     | ML                       | Establishes a logistic relationship between EEG features and classification output.                           |
| Random Forest (RF)           | ML                       | An ensemble method that builds multiple decision trees for robust classification.                             |
| Support Vector Machine (SVM) | ML                       | Maps EEG features into higher-dimensional space to improve separation between classes.                        |
| XGBoost                      | ML                       | XGBoost is a fast, accurate model that predicts EEG-based conditions by learning patterns from brain signals. |

The above table 3.3 provides an overview of commonly used machine learning models for EEG signal classification. Each model employs a distinct approach to analyse EEG features and enhance classification accuracy.

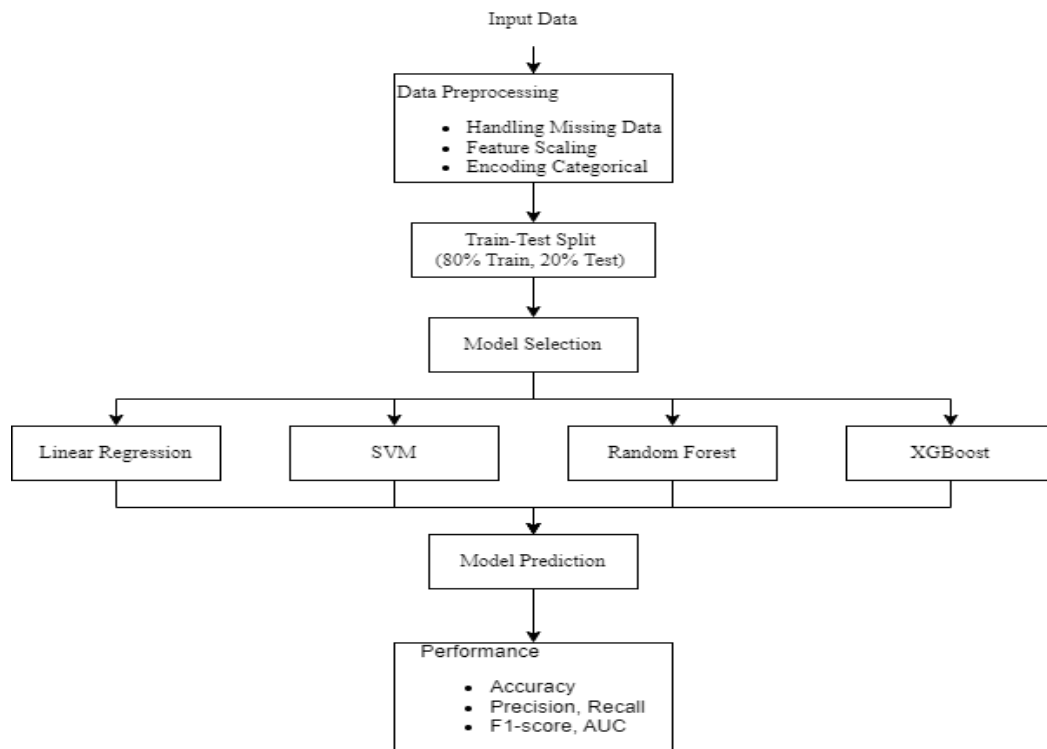


Fig no. 3.4 Model Architecture

In the above Fig 3.4, four machine learning models—Logistic Regression (LR), Support Vector Machine (SVM) and Random Forest (RF) were utilized to perform classification tasks. Each model employs a distinct methodology to learn from the data and generate predictions.

Logistic Regression (LR) is a linear classification algorithm that computes the optimal decision boundary by calculating a weighted sum of the input features. It applies the sigmoid function to transform the output into a probability between 0 and 1, classifying instances based on a predefined threshold (typically 0.5). Owing to its simplicity and interpretability, Logistic Regression is widely used for binary classification problems, particularly when the data exhibits a linear relationship.

Support Vector Machine (SVM) is a kernel-based algorithm designed to identify the optimal hyperplane that effectively separates data points from different classes. Unlike Logistic Regression, which relies on a linear boundary, SVM maximizes the margin between the nearest data points of each class (known as support vectors) to improve generalization. Additionally, SVM incorporates kernel functions (such as Radial Basis Function (RBF) and polynomial kernels) to handle non-linear data by mapping it into a higher-dimensional space, making linear separation possible.

XGBoost (Extreme Gradient Boosting) is an advanced boosting algorithm that builds an ensemble of decision trees sequentially, where each new tree focuses on correcting the errors made by the previous ones. It uses gradient boosting techniques to minimize loss functions efficiently, offering high performance and scalability.

XGBoost incorporates regularization to reduce overfitting and can handle missing data and complex feature interactions, making it particularly effective for large and noisy datasets.

Random Forest (RF) is an ensemble learning technique that constructs multiple decision trees using random subsets of the training data through bootstrap aggregation (bagging). Each decision tree is trained independently, and the final prediction is determined by aggregating the outputs, either through majority voting (for classification) or averaging (for regression). Random Forest improves robustness, reduces variance, and minimizes the risk of overfitting compared to individual decision trees, making it suitable for complex datasets with high dimensionality.

All models in this project were trained on the dataset and evaluated using various performance metrics, including accuracy, precision, recall, F1-score, and AUC-ROC, to assess their effectiveness in classification tasks. Each model offers distinct advantages depending on the dataset's characteristics, computational efficiency, and interpretability, making them applicable across a wide range of predictive modelling scenarios.

Logistic Regression (LR) is a statistical model that establishes a logistic relationship between EEG features and classification output. It is particularly useful for binary classification tasks where the goal is to predict the probability of a specific brain state or condition based on extracted features.

Random Forest (RF) is an ensemble learning method that constructs multiple decision trees and combines their predictions to improve robustness and accuracy. By aggregating outputs from various trees, it reduces overfitting and enhances generalization, making it well-suited for EEG classification tasks where complex patterns need to be recognized.

Support Vector Machine (SVM) is a machine learning algorithm that maps EEG features into a higher-dimensional space to maximize the separation between different classes. By using kernel functions, SVM effectively distinguishes complex EEG patterns, making it a powerful tool for classifying mental states and cognitive processes.

XGBoost (Extreme Gradient Boosting) is a powerful ensemble learning algorithm that builds sequential decision trees, where each tree corrects the errors of the previous ones. It efficiently handles high-dimensional EEG features, captures complex relationships, and provides accurate classification of brain states while minimizing overfitting through regularization techniques.

## **6. Model Training and Storage**

Once a suitable model is chosen, it is trained using labelled EEG datasets. Training involves feeding the model with input data and optimizing its parameters to minimize classification or prediction errors. After successful training, the model is stored for later use in real-time applications. The trained model is saved in a structured format (e.g., TensorFlow Saved Model, PyTorch model format) to enable easy deployment and integration into the prediction pipeline.

To facilitate model deployment and future inference, the trained models were serialized and stored using the pickle library. This allows the models to be loaded later for real-time predictions in the React-based frontend application without requiring retraining. Storing models ensures that the same trained classifiers can be used consistently, maintaining performance and reliability.

Each trained model was saved as a .pkl file using `pickle.dump()`, allowing for quick retrieval and deployment. Additionally, the scaler used for feature transformation was also stored to ensure consistency in preprocessing new data before inference. The stored model can be reloaded using `pickle.load()`, and new EEG data can be pre-processed and classified using the trained model. This approach enhances efficiency by reducing computational overhead and enabling real-time predictions in the deployed system.

By integrating different ML approaches, the proposed methodology ensures robust classification performance, effectively distinguishing MDD from healthy individuals based on EEG-derived features.

## **7. Frontend Application (React)**

To enhance usability, the final predictions and EEG data insights are integrated into a frontend application developed using React. This application provides a user-friendly interface for clinicians, researchers, and end-users to interact with the system. The interface allows users to upload EEG data, visualize processed signals, review model predictions, and interpret diagnostic insights. A well-designed frontend ensures accessibility and ease of use, improving the overall adoption of the system.

## **8. Final Prediction**

The trained model is used to make predictions on new EEG data. Given a new EEG signal, the system processes it using the trained model to classify or predict outcomes such as the presence of depressive symptoms, cognitive workload, or other neurological conditions. The prediction results can be presented in real-time or stored for further analysis by medical professionals.

### 3.2 Flow Chart

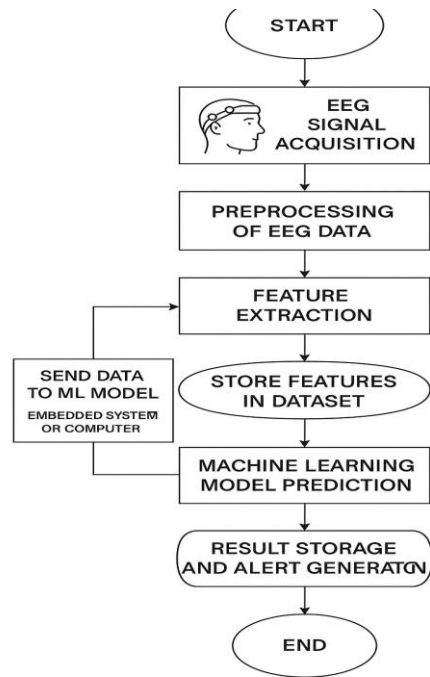


Fig no. 3.5 Flow chart of Proposed System

The diagram outlines a systematic process for improving the early detection of Major Depressive Disorder (MDD) using EEG signals and machine learning. The process begins with the acquisition of real-time EEG signals from subjects through an EEG headset, capturing brainwave activity that reflects mental states.

These raw EEG signals are then sent through a preprocessing stage, where noise, motion artifacts, and other irrelevant signals are removed. This ensures that only clean and usable data proceed to the next step. Once pre-processed, the signals undergo feature extraction, where meaningful characteristics—such as frequency bands, power spectral density, and entropy—are derived to represent the subject's mental condition.

The extracted features are stored in a dataset, forming a structured input for machine learning analysis. An embedded system or a connected computer plays a central role by transmitting this dataset to a trained machine learning model. This model is designed to classify the input data and predict the likelihood of the subject having symptoms of MDD.

Once the prediction is made, the results are stored, and if depressive patterns are detected, an alert is generated to notify medical professionals or connected health systems for further evaluation. This continuous cycle of data acquisition, analysis, and alerting supports early diagnosis and timely intervention.

### 3.3 Sequence Diagram

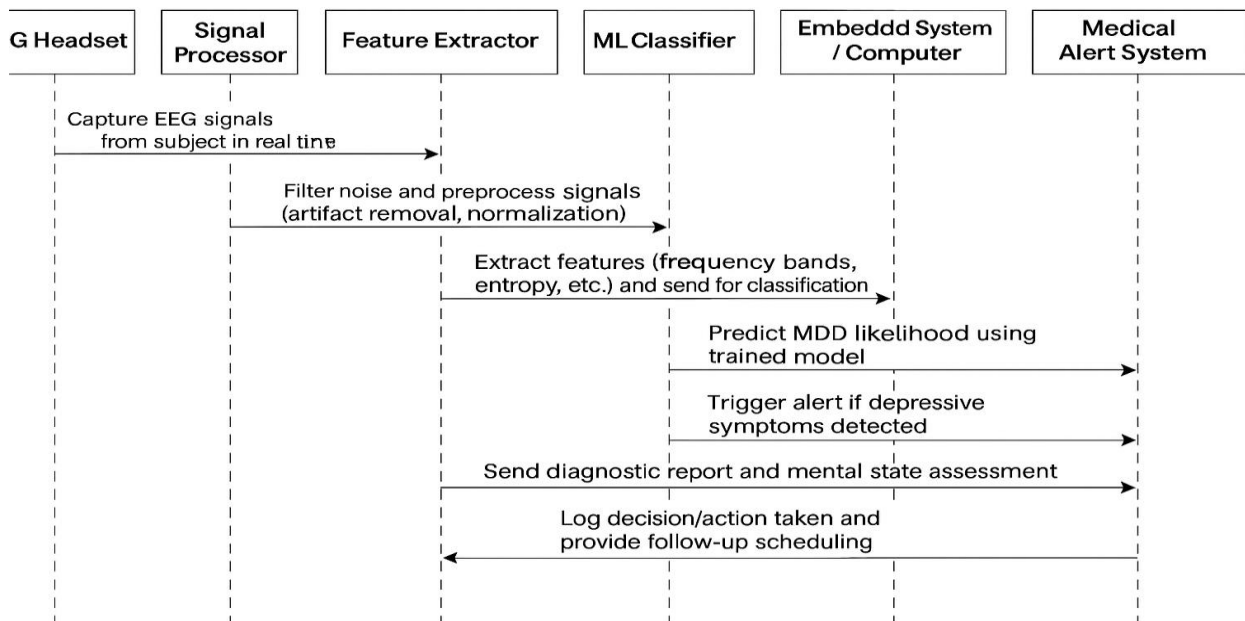


Fig no. 3.6: Sequence Diagram

The given sequence diagram represents an intelligent mental health monitoring system that utilizes EEG signal processing and machine learning to support the early detection of Major Depressive Disorder (MDD). The system comprises several interconnected components working in a coordinated manner to ensure timely and accurate analysis of brain activity.

First, the EEG headset captures real-time brainwave signals from the subject. These raw signals are sent to the Signal Processor, which filters out noise and artifacts (such as muscle movement or electrical interference) to ensure clean data is used for analysis. Once pre-processed, the data is passed to the Feature Extractor, which identifies significant neurological patterns such as frequency bands, entropy, or power spectral density that are indicative of the user's mental state.

These extracted features are then transmitted to the Machine Learning Classifier, which evaluates the input using a pre-trained model designed to recognize signs of depression. Based on this evaluation, the system determines the likelihood of the subject experiencing symptoms of MDD.

If depressive symptoms are detected, the Embedded System or Computer triggers an alert to the Medical Alert System. This ensures that healthcare providers are notified promptly. Simultaneously, a detailed report containing diagnostic insights and mental state assessments is sent to the Clinician Dashboard for review and medical decision-making.

Finally, the clinician logs the action taken, and the system may schedule follow-ups as needed. This intelligent health monitoring system leverages real-time bio signal analysis and AI-based classification to enable proactive mental healthcare, ensuring early intervention, improved outcomes, and continuous patient support.

### 3.4 ICA component map

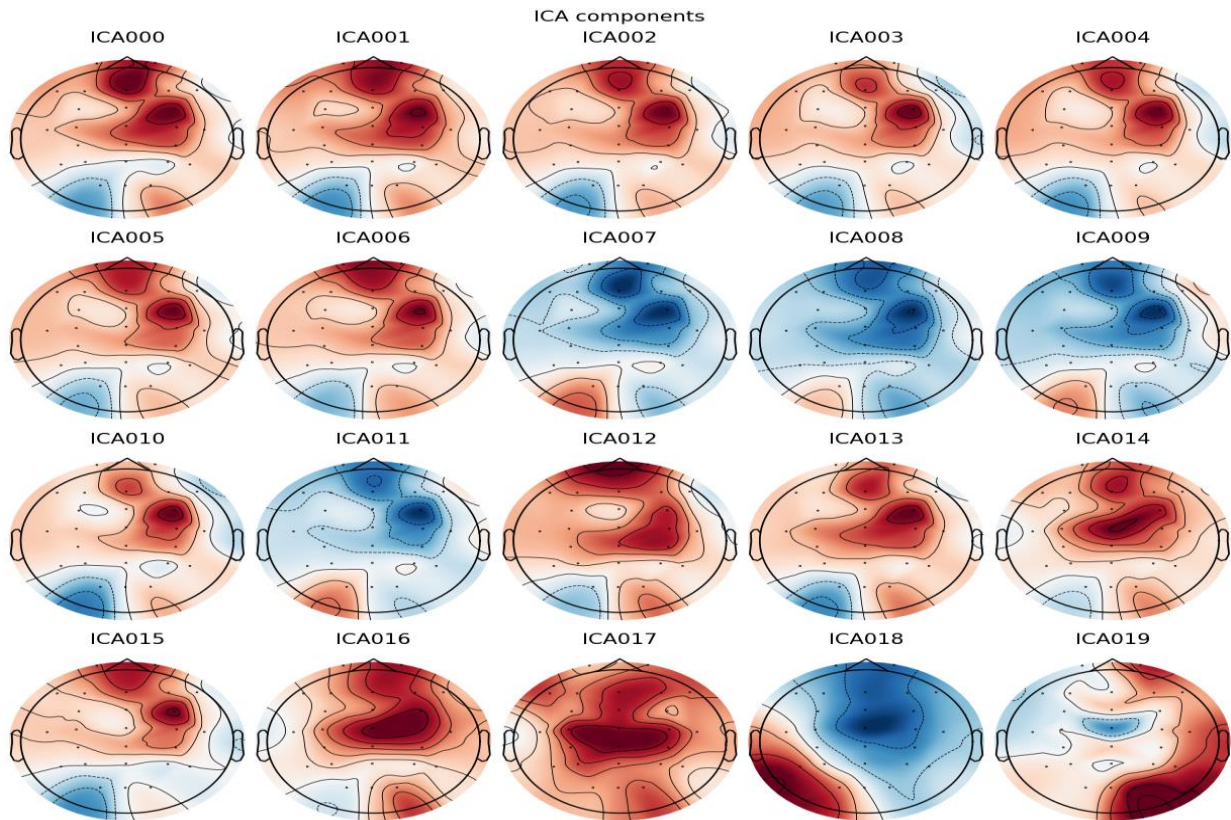


Fig no. 3.7 ICA component map

The above Fig 3.7 provides ICA (Independent Component Analysis) component map represents spatial topographies of EEG signal sources. Each circular plot shows an estimated independent component, with red and blue regions indicating positive and negative activations, respectively. These components help identify and remove artifacts like eye blinks and muscle movements from EEG data.

To mitigate powerline interference, a 50Hz notch filter is applied, followed by a bandpass filter (1–40 Hz) to eliminate low-frequency drifts and high-frequency muscle artifacts.

ICA works by decomposing the EEG signals into statistically independent components. These components are then analyzed to identify those corresponding to artifacts, which are removed before reconstructing the cleaned EEG signals. The identification of artifacts is performed manually or using automated algorithms based on predefined criteria such as spatial distribution, frequency characteristics, and correlation with external reference signals (e.g., EOG for eye movement detection).

### 3.5 Use Cases

- **Clinical Diagnosis Support:**
  - The system can assist psychologists and psychiatrists by providing objective EEG-based indicators of depressive symptoms, supporting more accurate and early diagnosis.
- **Home-Based Mental Health Tracking:**
  - Individuals can wear EEG headsets at home for continuous monitoring, allowing clinicians to track mental health trends without requiring constant in-person visits.
- **Early Detection in High-Risk Populations:**
  - The system can be used to screen high-risk groups—such as adolescents, postpartum women, or individuals with a family history of depression—for early warning signs before full symptoms develop.
- **Monitoring Treatment Effectiveness:**
  - Clinicians can use EEG data over time to evaluate the effectiveness of prescribed therapies (e.g., CBT, medication), enabling data-driven adjustments to treatment plans.
- **Employee Wellness Monitoring:**
  - Organizations can integrate the system into employee wellness initiatives to identify early stress and depression markers, promoting mental well-being in high-pressure industries
- **Integration with Hospital Systems:**
  - The system can be connected to broader smart healthcare infrastructures, feeding real-time mental health data into electronic health records for comprehensive care management.